5.5.2 Integration by Substitution (Definite Integrals)

Theorem: If g' is continuous on $[a,b]$ and *f* is continuous on the range of $u=g(x)$, then

$$\int_a^b f(g(x))g'(x)\,dx=\int_{g(a)}^{g(b)} f(u)\,du$$

Note: We see from the above theorem that we have to worry about changing the bounds when it comes to performing a u-substitution on definite integrals.

<u>Example 1:</u> Find $\int_0^{\sqrt{3}} 2x\sqrt{1+x^2}\,dx$.

We note $f(x)=2x\sqrt{1+x^2}$ is continuous on $\left[0,\sqrt{3}\right]$ so we can use FTCP2.

Let $u=1+x^2$ and thus $du=2xdx$

Our limits of integration become: $u(0)=1+(0)^2=1$ and $u\left(\sqrt{3}\right)=1+\left(\sqrt{3}\right)^2=4$

Therefore,

$$\int_0^{\sqrt{3}} 2x\sqrt{1+x^2}\,dx=\int_1^4 \sqrt{u}\,du=\int_1^4 u^{1/2}du=\frac{2}{3}u^{3/2}\Big|_1^4=\frac{2}{3}(4)^{3/2}-\frac{2}{3}(1)^{3/2}=\frac{2}{3}(8)-\frac{2}{3}(1)=\frac{16}{3}-\frac{2}{3}=\frac{14}{3}$$

Note: There are some differences between the u-substitutions in indefinite integrals and definite integrals that are worth noting. For indefinite integrals we always have to plug our original variable of integration back in for u after the integration is complete, and we need to remember to add a constant to our antiderivative.
On the other hand, for definite integrals we need to change the bounds, but we will NEVER plug back in for our original variable of integration after changing the bounds, and we need not add a constant to our antiderivative since any antiderivative will give the correct value of the definite integral by the FTCP2.

<u>Example 2:</u> Find $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \cot^3(x)\csc^2(x)\,dx$

We note $f(x)=\cot^3(x)\csc^2(x)$ is continuous on $\left[\frac{\pi}{4},\frac{\pi}{3}\right]$ so we can use FTCP2.

Let $u=\cot(x)$ and thus $du=-\csc^2(x)dx \;\;\Rightarrow\;\; -du=\csc^2(x)dx$

Our limits of integration become: $u\left(\frac{\pi}{4}\right)=\cot\left(\frac{\pi}{4}\right)=1$ and $u\left(\frac{\pi}{3}\right)=\cot\left(\frac{\pi}{3}\right)=\frac{1}{\sqrt{3}}$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} (\cot(x))^3\csc^2(x)\,dx=-\int_1^{1/\sqrt{3}} u^3du=-\left(\frac{1}{4}u^4\right)\Bigg|_1^{1/\sqrt{3}}=-\frac{1}{4}\left(\frac{1}{\sqrt{3}}\right)^4+\frac{1}{4}(1)^4=-\frac{1}{4}\left(\frac{1}{9}\right)+\frac{1}{4}=-\frac{1}{36}+\frac{1}{4}=\frac{8}{36}=\frac{2}{9}$$

Example 3: Determine each of the following.

a) Find $\displaystyle\int_1^2 \frac{dx}{(3-5x)^2}$.

We note $f(x)=\dfrac{1}{(3-5x)^2}$ is continuous on $[1,2]$ so we can use FTCP2.

Let $u=3-5x$ and thus $du=-5dx \;\Rightarrow\; -\dfrac{1}{5}du=dx$

Our limits of integration become: $u(1)=3-5(1)=-2$ and $u(2)=3-5(2)=-7$

Therefore,

$$\int_1^2 \frac{dx}{(3-5x)^2}=\int_{-2}^{-7}\frac{1}{u^2}\cdot\frac{-1}{5}du=-\frac{1}{5}\int_{-2}^{-7}u^{-2}du=\frac{1}{5}u^{-1}\Big|_{-2}^{-7}=\frac{1}{5}(-7)^{-1}-\frac{1}{5}(-2)^{-1}=-\frac{1}{35}+\frac{1}{10}=\frac{1}{14}$$

b) Find $\displaystyle\int_1^e \frac{\ln(x)}{x}dx$

We note $f(x)=\dfrac{\ln(x)}{x}$ is continuous on $[1,e]$ so we can use FTCP2.

Let $u=\ln(x)$ and thus $du=\dfrac{1}{x}dx$

Our limits of integration become: $u(1)=\ln(1)=0$ and $u(e)=\ln(e)=1$

Therefore,

$$\int_1^e \frac{\ln(x)}{x}dx=\int_0^1 u\,du=\frac{1}{2}u^2\Big|_0^1=\frac{1}{2}(1)^2-\frac{1}{2}(0)^2=\frac{1}{2}-0=\frac{1}{2}$$

c) Find $\displaystyle\int_0^1 e^{\left(e^x+x\right)}dx$

We note $f(x)=e^{\left(e^x+x\right)}$ is continuous on $[0,1]$ so we can use FTCP2. Also $\displaystyle\int_0^1 e^{\left(e^x+x\right)}dx=\int_0^1\left[e^{e^x}\cdot e^x\right]dx$.

Let $u=e^x$ and thus $du=e^x dx$

Our limits of integration become: $u(0)=e^0=1$ and $u(1)=e^1=e$

Therefore,

$$\int_0^1 e^{\left(e^x+x\right)}dx=\int_0^1\left[e^{e^x}\cdot e^x\right]dx=\int_1^e e^u du=e^u\Big|_1^e=e^e-e$$

Symmetry & Integrals

Theorem (Integrals of Symmetric Functions):
Suppose *f* is continuous on $\left[-a,a\right]$.

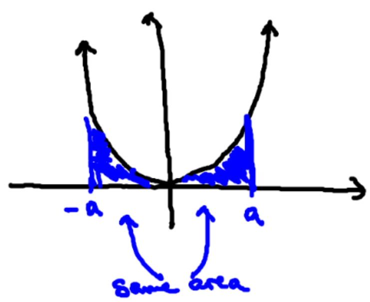
- If *f* is even $\left[f(-x)=f(x)\right]$, then $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$.
- If *f* is odd $\left[f(-x)=-f(x)\right]$, then $\int_{-a}^{a} f(x)\,dx = 0$.

General Idea: Consider the two following example functions which illustrate the above theorem.

Even: $f(x)=x^2$

We now this function is even because
$f(-x)=(-x)^2=x^2=f(x)$

We then see that $\int_{-a}^{a} x^2dx = 2\int_{0}^{a} x^2dx$.



Odd: $f(x)=x^3$

We now this function is odd because
$f(-x)=(-x)^3=-x^3=-f(x)$

We then see that $\int_{-a}^{a} x^3dx = 0$.

Proof:

We first note that $\int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{0}^{a} f(x)\,dx$

We perform a substitution on the integral $\int_{-a}^{0} f(x)\,dx$.

Let $u=-x$, then $du=-dx \;\Rightarrow\; -du=dx$ and $x=-u$.
Our limits of integration become: $u(-a)=-(-a)=a$ and $u(0)=-(0)=0$

Therefore we can say that $\int_{-a}^{0} f(x)\,dx = -\int_{a}^{0} f(-u)\,du = \int_{0}^{a} f(-u)\,du$.

So we have that $\int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{0}^{a} f(x)\,dx = \int_{0}^{a} f(-u)\,du + \int_{0}^{a} f(x)\,dx$.

- If *f* is even then $\int_{0}^{a} f(-u)\,du + \int_{0}^{a} f(x)\,dx = \int_{0}^{a} f(u)\,du + \int_{0}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$
- If *f* is odd then $\int_{0}^{a} f(-u)\,du + \int_{0}^{a} f(x)\,dx = -\int_{0}^{a} f(u)\,du + \int_{0}^{a} f(x)\,dx = 0$

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Example 4: Find $\int_{-2}^{2}\left(x^6+1\right)dx$.

We first notice that $f\left(x\right)=x^6+1$ continuous on $\left[-2,2\right]$ so we can use the FTCP2.

We also see that *f* is an even function because: $f\left(-x\right)=\left(-x\right)^6+1=x^6+1=f\left(x\right)$

Therefore we compute

$$\int_{-2}^{2}\left(x^6+1\right)dx=2\int_{0}^{2}\left(x^6+1\right)dx=2\left(\frac{1}{7}x^7+x\right)\Bigg|_0^2=2\left[\frac{1}{7}(2)^7+(2)\right]-0=\frac{256}{7}+4=\frac{284}{7}$$

Note: This integral could have still be evaluated without making use of the fact that $f\left(x\right)$ was an even function. However, at times evaluating integrals in this way can be easier since one of the limits of integration becomes 0.

Example 5: Find $\displaystyle\int_{-1}^{1}\frac{\tan\left(x\right)}{1+x^2+x^4}dx$.

We note that $f\left(x\right)=\dfrac{\tan\left(x\right)}{1+x^2+x^4}$ is continuous on the interval $\left[-1,1\right]$ so we can use the FTCP2.

We also see that *f* is an odd function because: $f\left(-x\right)=\dfrac{\tan\left(-x\right)}{1+\left(-x\right)^2+\left(-x\right)^4}=\dfrac{-\tan\left(x\right)}{1+x^2+x^4}=-f\left(x\right)$

Therefore we compute $\displaystyle\int_{-1}^{1}\frac{\tan\left(x\right)}{1+x^2+x^4}dx=0$.

Note: We can see that being able to determine that a function is odd can make the evaluation of certain integrals extremely easy.